

Python Web Development

6-Week Intensive Curriculum

From Fundamentals to Full-Stack Deployment

WEEK 1 — Python Fundamentals

Submodule 1: Introduction to Python

- Installing Python
- print function
- Comments
- Input & Output

Submodule 2: Variables, Data Types & Operators

- Variables
- Data types and its methods
- Operators and types

Submodule 3: Conditions, Loops & Strings

- if-else & elif ladder
- for & while loops
- String methods & f-strings

Submodule 4: Data Structures

- Lists & list methods
- Tuples, sets and Dictionaries
- Stack and queue

PRACTICE PROJECTS

- Smart Calculator
- Digital Contact Vault

WEEK 2 — File Handling & Object-Oriented Programming

Submodule 1: Functions

- Creating functions
- Parameters & arguments
- Higher order functions

Submodule 2: File Handling

- Reading files
- Writing files
- Append mode
- Working with text files

Submodule 3: Exception Handling

- try-except
- finally block
- Custom error messages

Submodule 4: Object-Oriented Programming

- Classes & objects
- Constructors
- Inheritance
- Encapsulation

PRACTICE PROJECTS

- Campus Record Manager
 - Student Management System
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WEEK 3 — Frontend Basics (HTML & CSS)

Submodule 1: HTML Basics

- HTML structure
- Headings & paragraphs
- Links & images
- Lists, tables & forms

Submodule 2: CSS Basics

- CSS syntax & selectors
- Colors & backgrounds
- Margin, padding & borders
- Fonts & text styling

Submodule 3: Layout & Responsive Design

- Flexbox basics
- Grid basics
- Media queries

Submodule 4: CSS Frameworks & Integration

- Bootstrap basics
- Tailwind CSS basics
- Responsive components
- Connecting HTML & CSS

PRACTICE PROJECTS

- Developer Portfolio Website
 - Coffee Shop Landing Page
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WEEK 4 — Flask & Database Development

Submodule 1: Database Basics

- Introduction to databases
- SQL basics
- CRUD operations
- SQLite basics

Submodule 2: Flask Introduction

- Flask setup
- Routes
- Templates
- Static files

Submodule 3: Flask Forms & CRUD

- HTML forms
- Form handling
- CRUD application
- Request methods

Submodule 4: SQLAlchemy ORM Basics

- SQLAlchemy setup
- Models
- Database queries
- ORM relationships basics

PRACTICE PROJECTS

- Secure Notes Hub

WEEK 5 — Advanced Flask & Security

Submodule 1: Authentication System

- Login & registration
- Password hashing
- Sessions
- Logout functionality

Submodule 2: Flask Project Features

- Flash messages
- File uploads
- Search functionality
- Pagination basics

Submodule 3: Project Structure & Optimization

- Organizing files
- Blueprint basics
- Reusable templates
- Code cleanup

PRACTICE PROJECTS

- TaskFlow Manager
- StudentSphere Portal

WEEK 6 — Testing, Deployment & Final Project

Submodule 1: Git & GitHub

- Git and GitHub basics
- Push & pull changes
- Branch basics

Submodule 2: Testing & Debugging

- Error handling

- Common Flask errors
- Testing basics

Submodule 3: Deployment Basics

- Hosting basics
- Environment variables
- Deploying Flask projects

FINAL PROJECT IDEAS

- SkillSync Platform
 - NoteNest Application
 - BookVerse System
 - WorkHive Manager
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